

DEVA ANAND

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Available for Fall 2026 internship: September 14 – December 30, 2026 (full-time)

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published NLP method into a self-serve embedding-and-scoring service for researchers.
- Built a YAML model registry, a versioned append-only construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system with deterministic language detection; grew the backend suite to **40 hermetic tests** that run in CI without ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) serving **8+ UK clinical clients** and **10,000+ monthly triage submissions** through REST integrations with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Architected a full-stack training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, building the pipeline that generated labeled training examples and cutting manual annotation by **100+ hours/month** while lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable and reliable.
- Owned backend security and reliability across the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Shipped a production LLM auto-commenting service using **4-bit quantized Llama-7B** via llama.cpp on a single VM, with async queued generation, per-character rate limits, short-circuit caching, and a deterministic content-safety filter.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** NYC TLC dataset by swapping shuffle joins for broadcast joins (minutes to ~0.05s), enabling AQE and skew-join handling, and tuning shuffle partitions, schema-aware column-pruned reads, and year/month partitioning.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with **perf** and inspecting generated assembly to localize memory-hierarchy bottlenecks.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval and ranking component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, Java, Go, TypeScript, JavaScript, SQL, Bash

Systems & Data: Spark / PySpark, Hadoop, Kafka, distributed systems, load balancing, Linux perf, SIMD, OpenMP

Backend: Node.js, Express, FastAPI, REST APIs, async queues, MongoDB, PostgreSQL, Redis, DynamoDB

ML & Retrieval: PyTorch, sentence-transformers, embeddings, RRF, cross-encoder reranking, MMR, RAG, llama.cpp

Cloud & DevOps: AWS, GCP, Azure, Kubernetes, Docker, Git, GitHub Actions, CI/CD